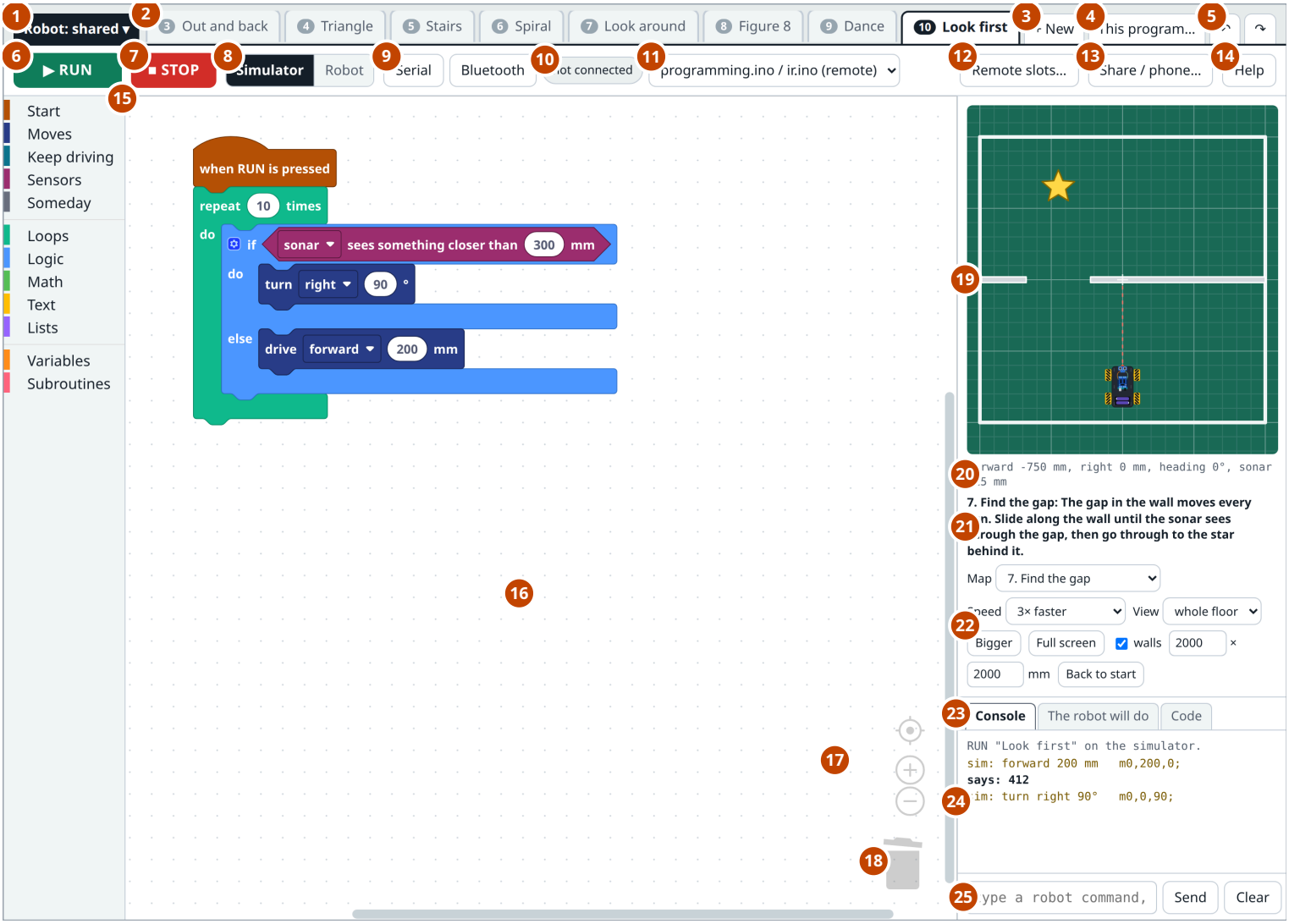
The Blockly screen

What everything is and what it does.

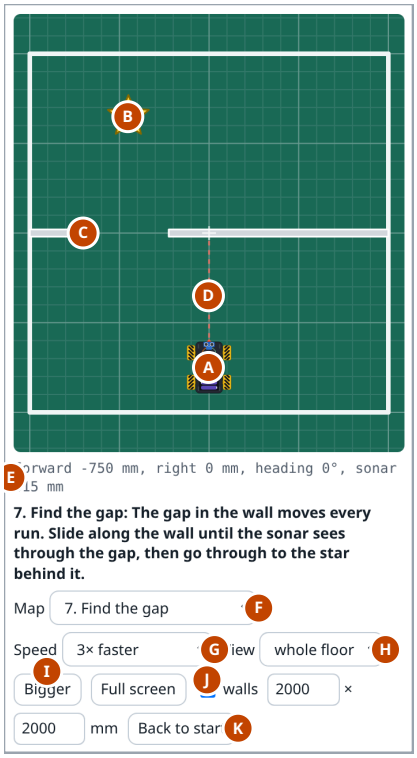


1. **Robot** Whose tabs these are. Leave it on *shared*. Connecting a robot never changes your tabs.
2. **Program tabs** Each tab is one program. Click one to open it. Tabs 1 to 9 are also the remote's slots 1 to 9.
3. **+ New** A blank program, a copy of a demo, an example (like *Look first*), or the *Toolbox* of subroutines.
4. **This program...** Rename, duplicate or delete the open program.
5. **Undo, redo** Take back the last change (Ctrl+Z), or put it back.
6. **RUN** Runs the program in the open tab, on the simulator or on the robot.
7. **STOP** Stops the program and the wheels. Always works.
8. **Run on** *Simulator*: the pretend robot on the right. *Robot*: your real robot, once connected.
9. **Serial, Bluetooth** Connect to your robot: *Serial* with a USB cable, *Bluetooth* without one.
10. **Connection** Says *not connected* or how the robot is connected.
11. **Sketch** Which program is on the robot's Arduino. It picks itself when the robot connects.
12. **Remote slots...** Turns the program into a remote slot, sends it to the robot, or reads the robot's slots back.
13. **Share / phone...** A link, a QR code, the program as text or a file. *Bring a program in* pastes one back.
14. **Help** What the buttons and blocks do.
15. **Categories** Click one to see its blocks, then drag a block out onto the workspace.
16. **Workspace** Build the program here. Only blocks under *when RUN is pressed* run. Drag empty space to move around.
17. **Center, zoom** Bring the blocks back into view, make them bigger or smaller.
18. **Trash** Drag a block here to delete it (or select it and press Delete).
19. **Simulator** The pretend robot on a floor, 10 cm squares. Drag the robot to choose where it starts.
20. **Readout** Where the robot is, which way it faces, and what the sonar sees.
21. **Map goal** What to do on this map. Reach the star to see *Made it!*
22. **Simulator controls** Pick a *Map*, the *Speed*, the *View*, a bigger picture, and *Back to start*.
23. **Console, The robot will do, Code** What happened, the list of moves your program makes, and the program as code.
24. **Console** Every move, every *say* block, and every message from the robot.
25. **Command box** Type a robot command (like *u* for the sonar) and press Send. *Clear* empties the console.

A closer look

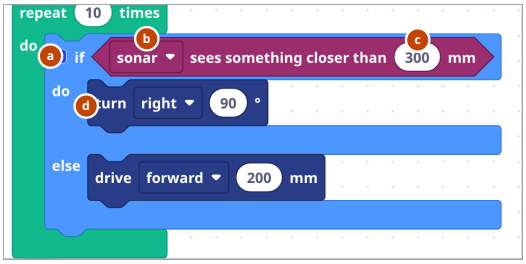
The simulator, a block and its menu, and the console.

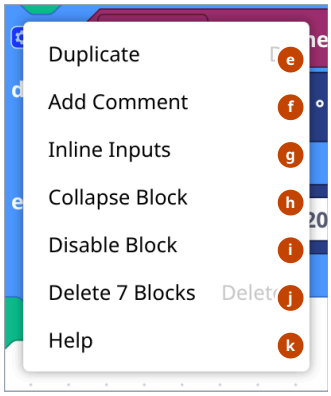
The simulator



- **A** **The robot** Drag it to choose where it starts. RUN and *Back to start* put it there. The orange arrow is its nose.
- **B** **The star** The goal. Reach it and the console says *Made it!* with your time and bumps.
- **C** **Walls** The sonar sees them and the robot bumps into them. On some maps they move, or open and close.
- **D** **Dashed line** What the sonar sees: straight ahead, to the nearest wall.
- **E** **Readout** Position, heading and the sonar reading. Watch it while the program runs.
- **F** **Map** Ten floors, from easy to hard. From map 4 on, something changes every run.
- **G** **Speed** *real time* to watch closely, faster to finish quickly.
- **H** **View** The whole floor, or close up around the robot.
- **I** **Bigger, Full screen** A bigger simulator, or the whole screen (Esc to leave).
- **J** **walls, size** The room's walls on or off, and its size in mm.
- **K** **Back to start** Puts the robot back where it starts (press STOP first).

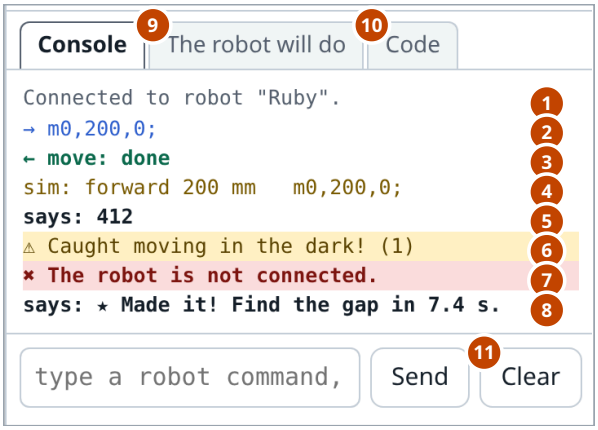
A block, and its right-click menu





- **a** **Gear** Changes the block: here it adds *else if* and *else*. Subroutines use it to add inputs.
- **b** **Dropdown** Click the little triangle to choose: *sonar* or *laser*, *right* or *left*.
- **c** **Number** Click it and type. You can also drop a variable or a math block on it.
- **d** **Snap** Blocks snap together top to bottom. The ones inside *do* and *else* are what *if* does.
- **e-k** **Right-click a block** for this menu:
  - **e** **Duplicate** a copy, right next to it
  - **f** **Add Comment** a note for people (the program ignores it)
  - **g** **Inline Inputs** the inputs in one row or stacked
  - **h** **Collapse Block** fold it up small (Expand opens it again)
  - **i** **Disable Block** the program skips it for now
  - **j** **Delete** the block and everything inside it
  - **k** **Help** about this kind of block

The console



- **1** **plain** A message from the page: what it is doing.
- **2** → A command the page sent to the robot.
- **3** ← What the robot answered.
- **4** **sim:** What the pretend robot did.
- **5** **says:** A *say* block in your program.
- **6** ⚠ A warning: something to look at.
- **7** ✖ An error: the program stopped.
- **8** ★ **Made it!** You reached the star.
- **9** **The robot will do** The moves your program will make, worked out before it runs.
- **10** **Code** Your blocks as JavaScript code.
- **11** **Clear** Empties the console (or use the *clear the console* block).

Copy and paste: click a block, Ctrl+C, then Ctrl+V. It works from one tab to another too.

Lost your blocks? Press the center button (17 on page 1), or drag the empty workspace.

Saved by itself: your tabs stay in this browser. To keep a copy, use *Share / phone...*, *Save as a file*.